

Project Navi

Bachelor Designer & Developer

NOAILLES Thomas



PROJECT NAVI

Software Requirements Specification

Company Presentation

Project Owner

This project is developed as part of a Bachelor. It is an individual academic project carried out by the student under the supervision of the teaching staff. The project owner is responsible for the planning, analysis, design, development, testing, and documentation of the application throughout the project lifecycle.

Academic Context

The Mobile Navigation Assistant project is developed within the framework of a Bachelor's project. Its objective is to apply software engineering methodologies to the design and development of a mobile application that addresses a real-world accessibility challenge. The project follows a structured development process, including opportunity analysis, project planning, feasibility study, risk analysis, functional analysis, requirements specification, software design, implementation, and testing.

Project Organization

Role	Responsibility
Project Manager	Project planning and organization
Software Developer	Mobile application development
UI/UX Designer	User interface and accessibility design
Tester	Functional testing and validation
Academic Supervisor	Project monitoring and guidance

Although these responsibilities are traditionally assigned to different team members, they are all carried out by the project owner within the scope of this individual academic project.

Stakeholders

Stakeholder	Role
Project Owner	Project development and management
Academic Supervisor	Technical and academic guidance
Visually Impaired Users	End users and testers
Accessibility Organizations	Feedback and validation

The stakeholders contribute to the project by providing supervision, technical feedback and user validation. Their involvement ensures that the application addresses real user needs.

Project Presentation

Background

Visual impairment can make independent outdoor navigation challenging due to obstacles, pedestrian crossings, traffic lights, and constantly changing urban environments. Although traditional mobility aids such as white canes and guide dogs provide essential support, they do not always provide detailed information about the surrounding environment.

Recent advances in artificial intelligence and mobile technologies make it possible to develop applications capable of recognizing objects, interpreting the environment, and providing real-time audio guidance. This project aims to explore how these technologies can improve mobility and

independence for visually impaired users while complementing existing mobility aids.

Problem Statement

Visually impaired individuals often face difficulties when navigating unfamiliar environments. Existing navigation applications generally focus on route planning but provide limited assistance for detecting nearby obstacles or interpreting the immediate surroundings.

The challenge addressed by this project is to design a mobile application capable of assisting visually impaired users during outdoor navigation by combining GPS positioning, artificial intelligence, and voice guidance in a simple, accessible, and reliable interface.

Project Objectives

The primary objective of the project is to develop a mobile application that assists visually impaired users during outdoor navigation.

The application aims to:

- provide accurate GPS navigation;
- detect obstacles using artificial intelligence;
- recognize elements such as pedestrian crossings and traffic lights;
- provide real-time voice guidance;
- offer an accessible and user-friendly interface;
- improve user confidence and independence during travel.

Project Scope

The project focuses on the design and development of a mobile navigation assistant intended for outdoor use. The application integrates artificial intelligence, GPS positioning, and voice interaction to provide navigation support.

The project includes the analysis, design, implementation, and testing of the application prototype.

Features outside the project scope include autonomous navigation without user supervision, medical diagnosis, indoor navigation, and the replacement of traditional mobility aids such as white canes or guide dogs.

Target Audience

Primary Users

The primary target audience consists of visually impaired individuals who require assistance during outdoor navigation. The application is designed for users with varying degrees of visual impairment, ranging from low vision to complete blindness. Its purpose is to improve mobility, increase independence, and enhance safety while travelling.

Secondary Users

Secondary users include accessibility organizations, testers, and academic supervisors involved in the validation and evaluation of the project. Their feedback contributes to improving the application's usability, accessibility, and overall reliability.

User Needs

The application has been designed to address the following user needs:

- Safe outdoor navigation.
- Reliable obstacle detection.
- Real-time voice guidance.
- Simple and intuitive interaction.
- Accessible interface with high contrast and screen reader compatibility.
- Fast and reliable response.

Accessibility Considerations

Accessibility is a fundamental requirement of the project. The interface must remain simple, readable, and fully compatible with accessibility features available on modern smartphones. Voice interaction, high-contrast colors, large touch areas, and minimal user interaction during navigation are prioritized to provide an inclusive user experience.

Primary User Persona

The target audience was defined using user personas developed during the opportunity study. These personas helped identify user expectations, accessibility requirements, and common navigation challenges.

Lucas

age: 21

residence: Creteil

education: Bachelor student

occupation: Administrative Assistant

marital status: University Student



I want to explore new places without having to rely on someone else.

Lucas is a visually impaired university student who is very comfortable with technology. He uses accessibility features on his smartphone daily and relies on public transportation to move around independently. He enjoys discovering new places but often feels stressed when navigating unfamiliar environments.

Comfort With Technology

INTERNET



SOFTWARE



MOBILE APPS



SOCIAL NETWORK



Criteria For Success:

- Navigate independently in unfamiliar places.
- Avoid unexpected obstacles.
- Reach destinations safely and confidently.

Needs

- Real-time obstacle detection.
- Accurate voice guidance.
- Fast and intuitive interactions.

Values

- Independence.
- Efficiency.
- Innovation.
- Accessibility.

Wants

- Personalized navigation preferences.
- Integration with public transportation information.

Fears

- Missing important obstacles.
- Losing orientation in unfamiliar areas.
- Depending on others for mobility.

Competitive Analysis

Several mobile applications have been developed to support visually impaired users in their daily activities. These applications provide features such as object recognition, text reading, navigation assistance, or remote human assistance. The most relevant solutions identified during the opportunity study are presented below.

Existing Solutions

App	Features	Strengths	Limitations
Seeing AI	Object recognition, text reading, scene description	Accurate AI features	Limited navigation assistance
Google Lookout	Object detection, document reading, image recognition	Easy to use, Android integration	Focuses on object recognition rather than navigation
Be my eyes	Live assistance from volunteers	Human support for complex situations	Requires an Internet connection and volunteer availability

Strengths and Weaknesses

The benchmark highlights that current applications provide valuable accessibility features but often specialize in a limited set of tasks. Some focus on object recognition, while others rely on remote human assistance or basic GPS navigation. Few applications combine artificial intelligence, navigation, obstacle detection, and accessibility into a single solution designed specifically for independent outdoor mobility.

Project Positioning

The proposed Mobile Navigation Assistant aims to combine several complementary technologies within a single accessible application. Unlike existing solutions, the project focuses on integrating GPS navigation, artificial intelligence-based environmental analysis, and real-time voice guidance to support visually impaired users during outdoor travel. The application is intended to complement traditional mobility aids such as white canes or guide dogs rather than replace them.

Competitive Advantage

The proposed solution differentiates itself by combining artificial intelligence, GPS navigation, and accessibility features into a single application. The project emphasizes user safety, ease of use, and inclusive design while considering ethical and regulatory aspects related to artificial intelligence.

Graphic and Ergonomic Design

Design Principles

The graphical and ergonomic design of the Mobile Navigation Assistant is based on accessibility, simplicity, and usability principles. Since the application is intended for visually impaired users, every interface element has been designed to minimize cognitive effort while ensuring safe and intuitive interaction.

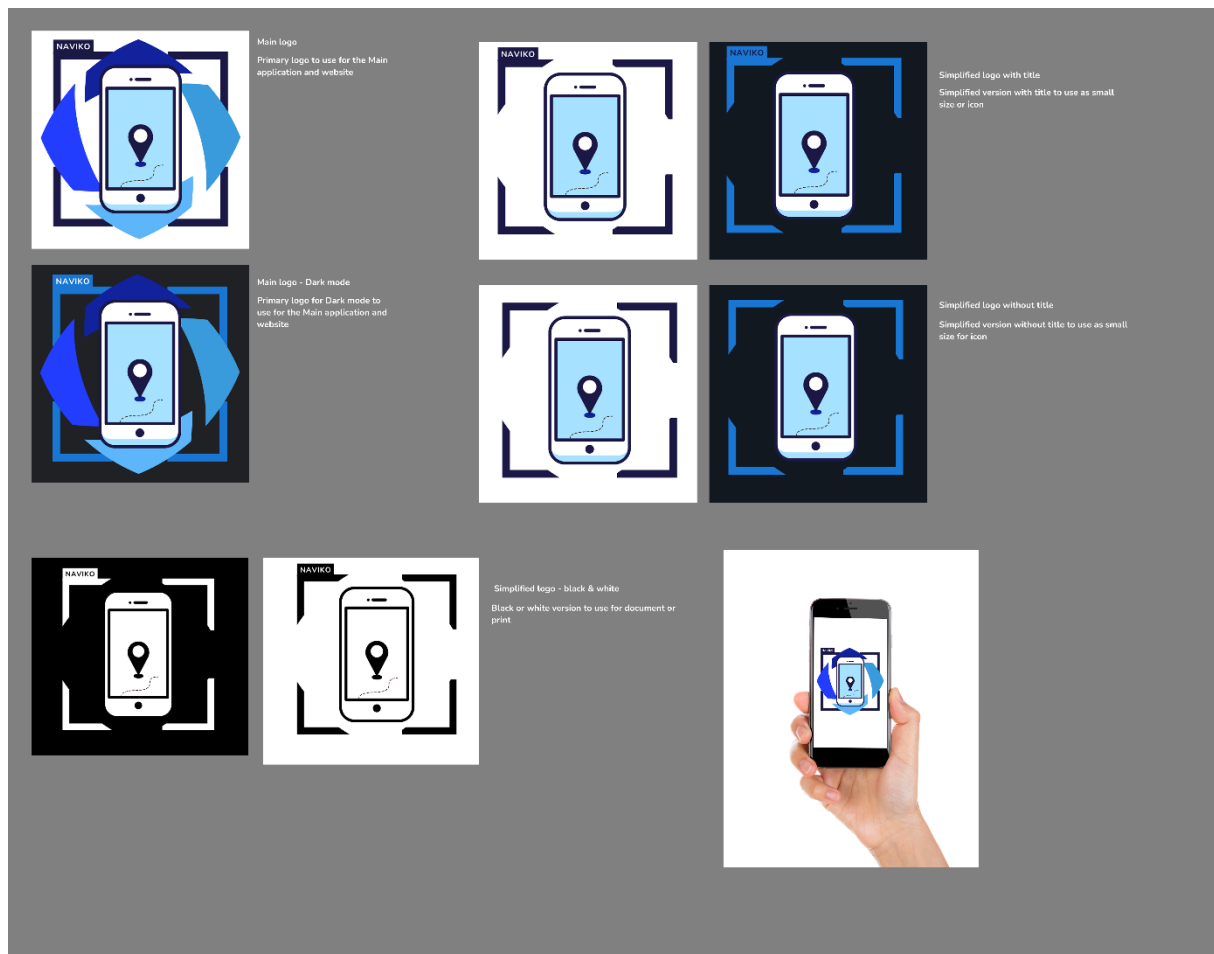
The interface prioritizes essential information, reduces visual clutter, and provides clear navigation through voice guidance and large interactive elements. The design follows accessibility recommendations to improve readability and facilitate user interaction under different environmental conditions.

Application Name – Naviko

The name Naviko combines *Navi*, derived from *Navigation*, with *Ko*, chosen for its short, friendly and memorable sound. The objective was to create a name that is easy to pronounce in different languages while evoking the idea of a companion that assists users during their journeys.

As an additional reference, the syllable *Ko* can also be associated with the Japanese word **doko** (どこ), meaning "where". Although this was not the primary reason behind the name, it reflects the application's purpose of helping users know where they are and navigate their surroundings.

Logos



The Naviko logo was designed to visually represent the application's core purpose: assisting visually impaired users by combining mobile navigation with AI-powered object detection. Each element of the logo reflects a specific feature of the application.

The smartphone represents the application's primary platform. Since Naviko is designed as a mobile application, this element immediately communicates its intended use.

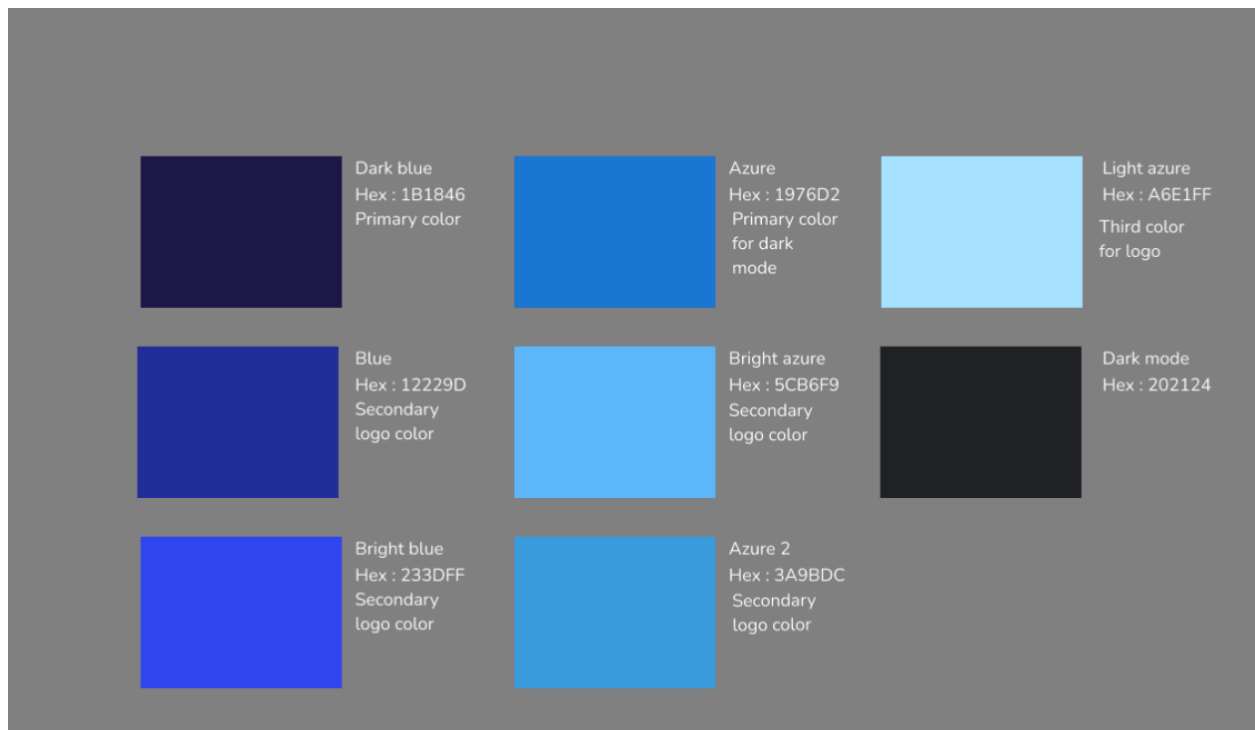
The location marker placed at the center symbolizes navigation, orientation and guidance. It represents the application's role in helping users move safely through their environment.

The surrounding geometric shapes evoke perception, guidance and the field of view used by the AI system. They also reinforce

the visual identity of the application while framing the smartphone.

The outer frame represents an AI detection bounding box, similar to those used in computer vision systems to identify objects. The application name Naviko is positioned where the detected object's label would normally appear, illustrating that the application itself acts as the intelligent detection system.

Color Palette

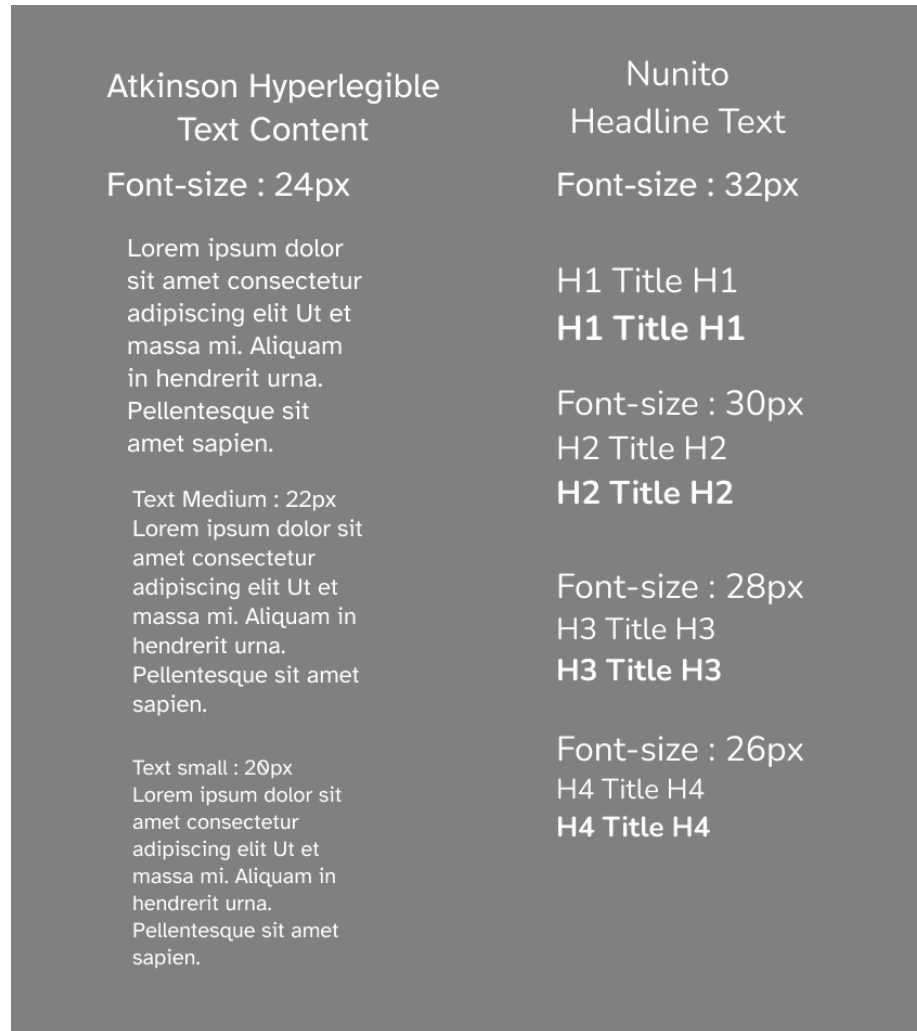




The interface primarily uses blue tones to inspire trust, reliability and accessibility. Orange is reserved for important actions requiring user attention, while green and red communicate positive and critical states respectively. Additional colors are

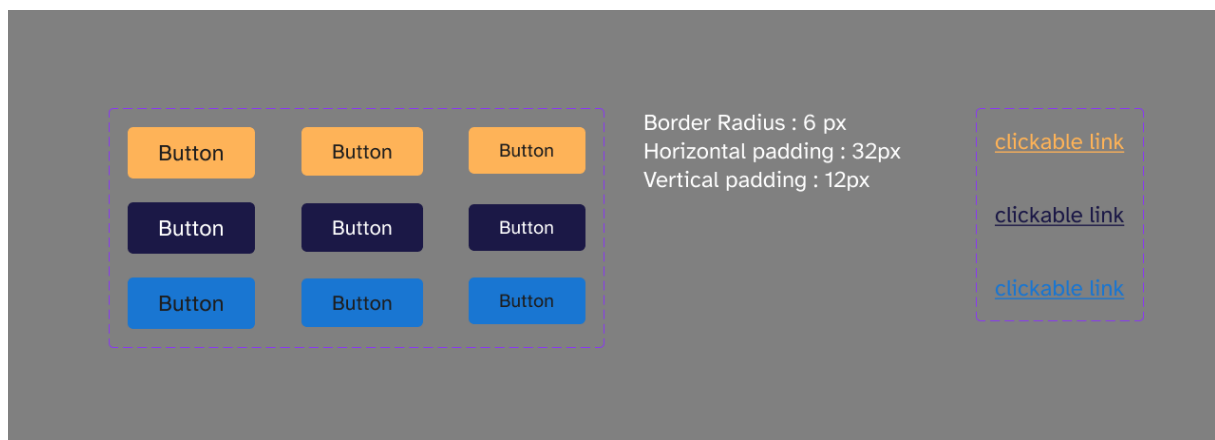
dedicated to AI object detection to provide clear visual feedback.

Typography



The typography combines accessibility and modern design. **Atkinson Hyperlegible** is used for body text due to its optimized readability for users with visual impairments, while **Nunito** is reserved for headings to establish a clear visual hierarchy and maintain a clean, contemporary appearance.

Components



Buttons are designed with high contrast and large clickable areas to improve accessibility. The primary button uses the application's primary color, while secondary buttons provide alternative actions without competing for the user's attention.

Icons



Icons follow a minimalist design to improve recognition and reduce cognitive load. Every important action is represented by both an icon and a text label whenever possible.

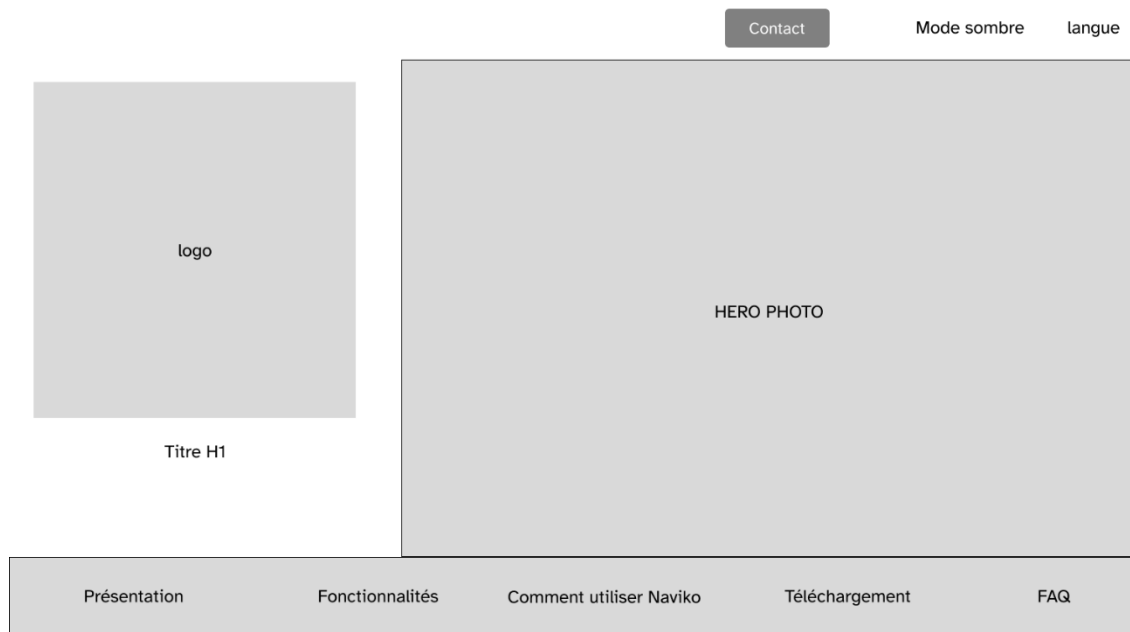
Photography

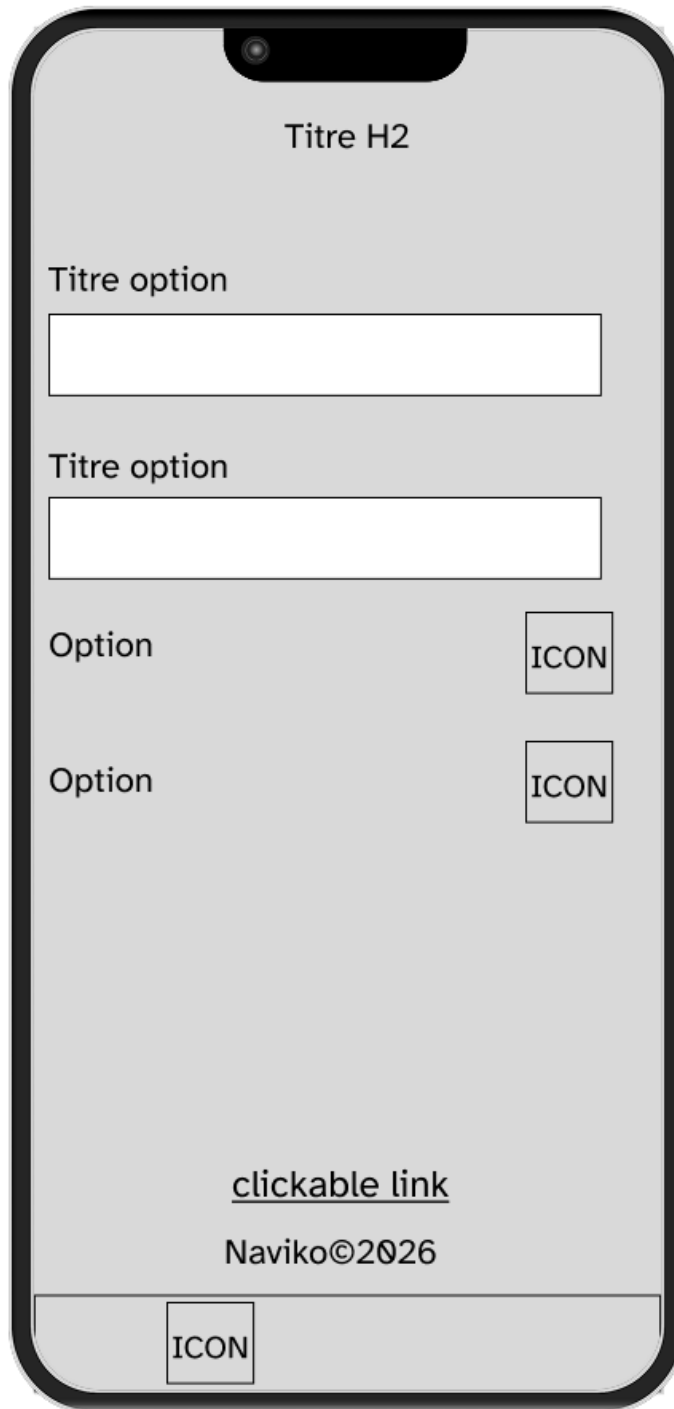
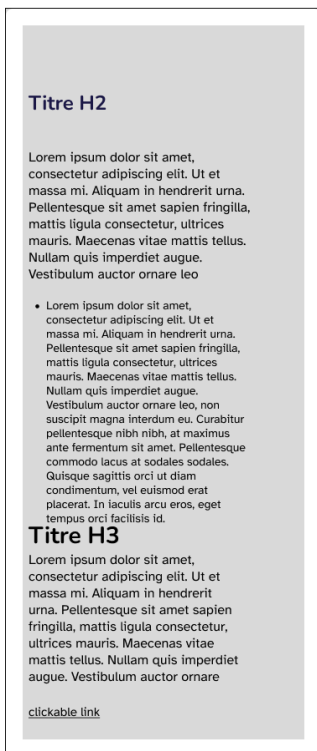
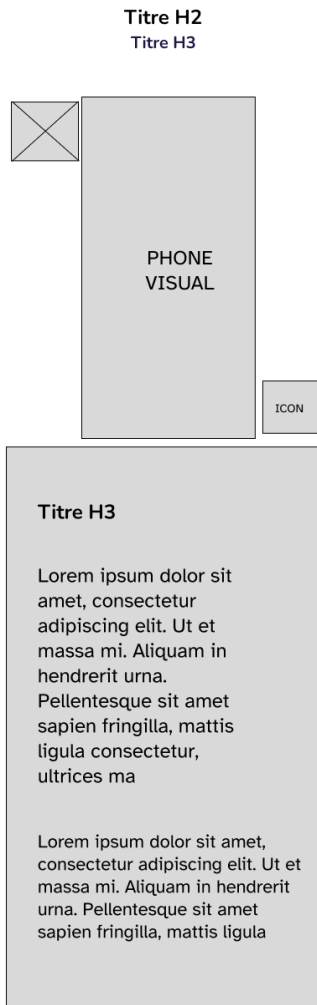
The application and website will use realistic, high-quality photographs depicting urban environments, pedestrian crossings, sidewalks and everyday navigation scenarios. Images will primarily feature situations that visually impaired users may encounter during their daily travels.

The objective is to clearly illustrate the application's features while maintaining a clean and professional visual identity.

Wireframes

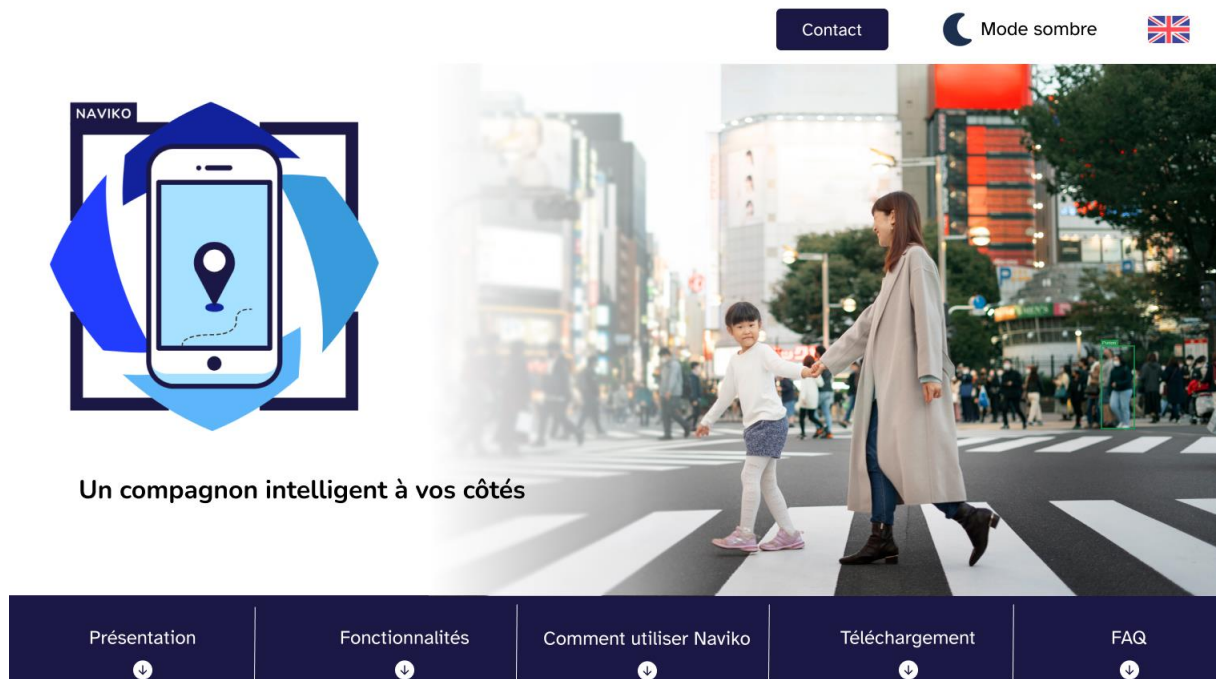
Low-fidelity wireframes are first developed to define the layout of each screen and the organization of interface elements. These preliminary designs focus on usability and navigation before the visual identity is applied.





High-Fidelity Mockups (Figma)

High-fidelity mockups are created using Figma to visualize the final appearance of the application. They integrate the selected color palette, typography, icons, and accessibility features while providing a realistic representation of the final user interface.



In addition to the mobile application, a lightweight promotional website will be developed. Its purpose is to introduce Naviko, present its features, answer common questions and provide contact information. The website does not replicate the application's functionality but serves as an information and communication platform.

Comment utiliser Naviko ?

Mode d'emploi

01



Première utilisation

Lors du premier lancement, Naviko vous demandera d'autoriser l'accès à l'appareil photo et d'accepter les conditions d'utilisation afin de vous accompagner lors de vos déplacements en toute sécurité.

* L'accès à l'appareil photo et l'acceptation des conditions d'utilisation sont requis pour utiliser les fonctionnalités de Naviko.



[Important] Utiliser Naviko en toute sécurité

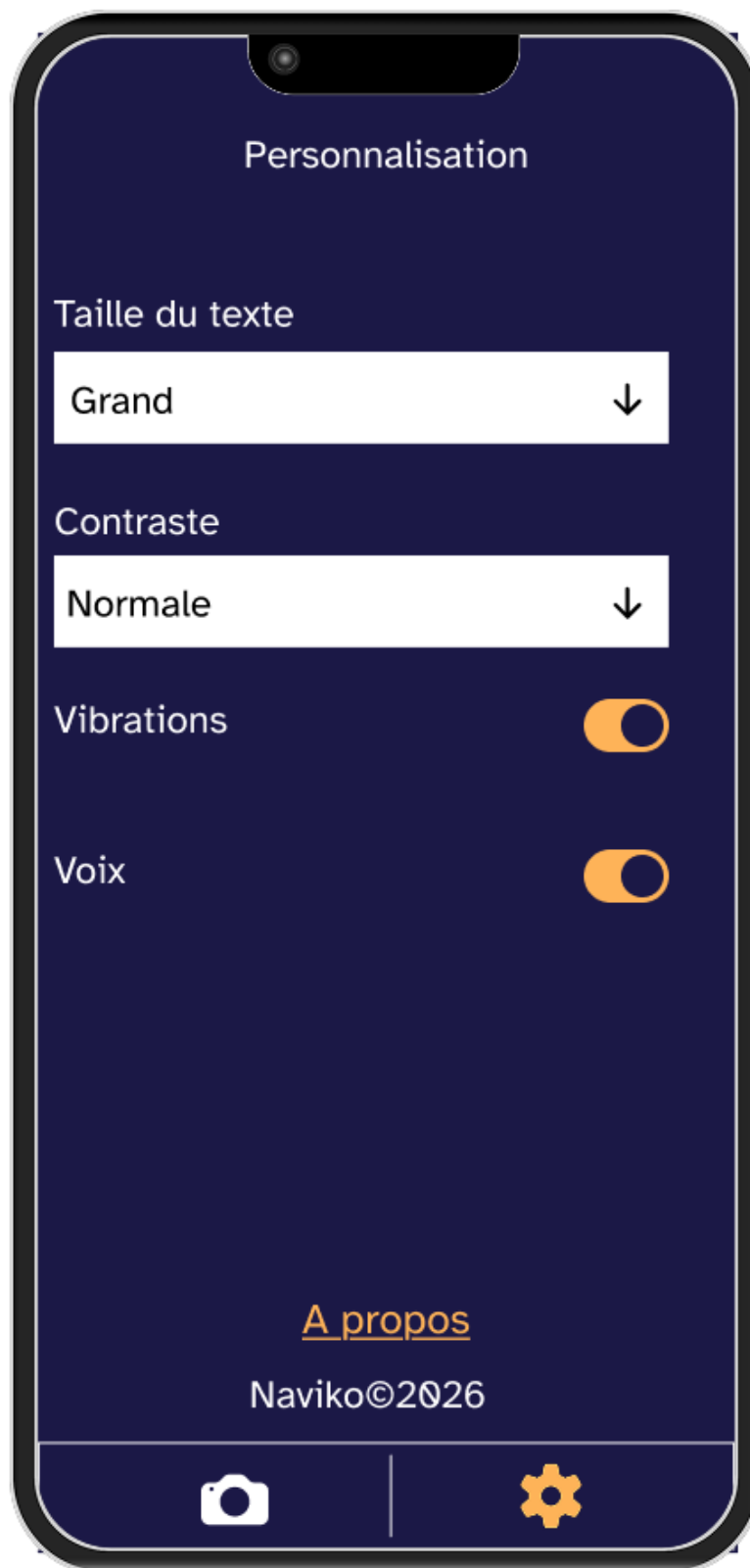
Naviko est conçu pour accompagner les personnes malvoyantes lors de leurs déplacements. L'application constitue une aide complémentaire et ne remplace ni la vigilance de l'utilisateur, ni une canne blanche, un chien guide ou tout autre moyen d'assistance.

- Orientez toujours la caméra vers l'avant pour garantir une analyse optimale.
- Utilisez l'application dans des conditions de visibilité suffisantes lorsque cela est possible.
- Vérifiez que la batterie de votre smartphone est suffisante avant un déplacement.
- Certaines détections peuvent être moins précises selon l'environnement, la météo ou les conditions d'éclairage.
- Gardez toujours le contrôle de vos déplacements et restez attentif à votre environnement.

A noter

Les performances de détection peuvent varier selon les conditions d'éclairage, la météo ou l'environnement. Naviko est conçu comme une aide complémentaire et ne remplace pas la vigilance de l'utilisateur ni les autres moyens d'assistance à la mobilité.

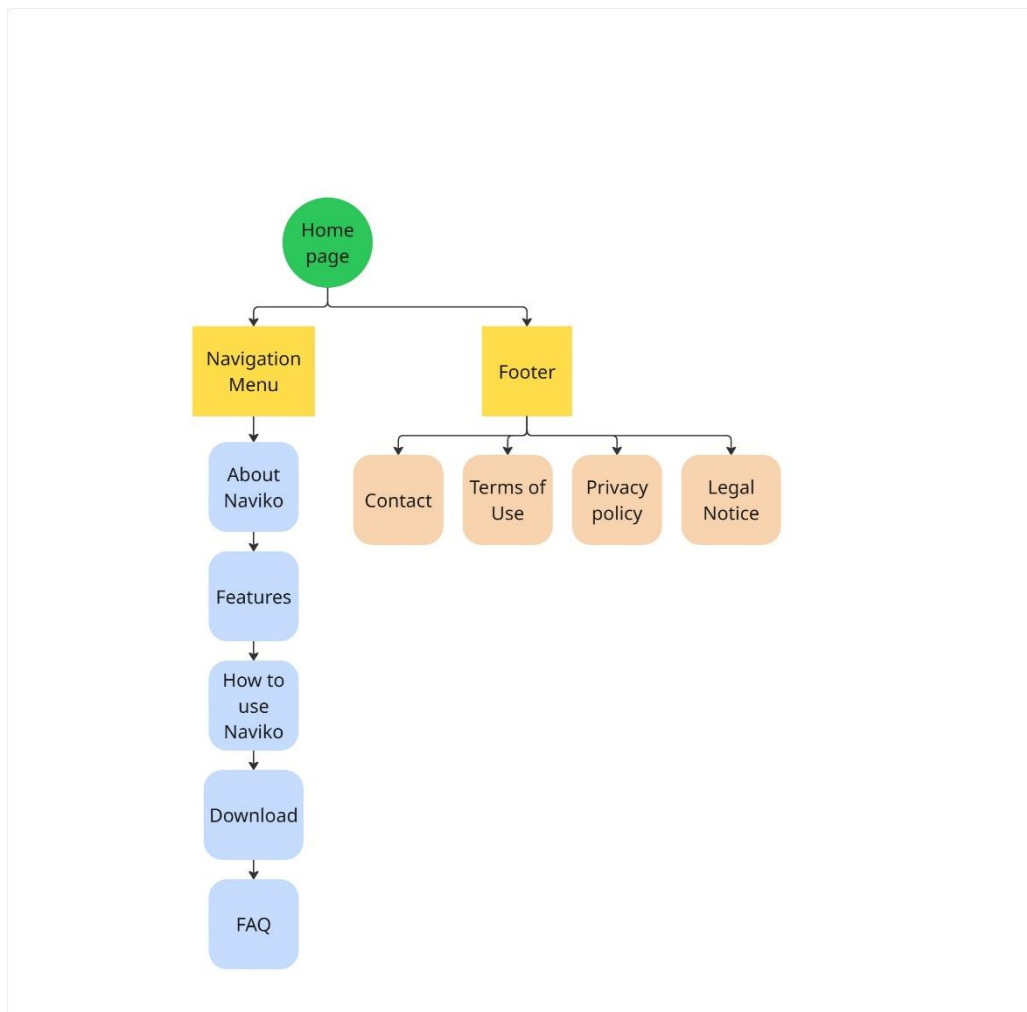
[Consultez nos Conditions d'utilisation](#)



Navigation Flow

The navigation flow illustrates how users interact with the different screens of the application. It has been designed to minimize the number of interactions required to access the main functionalities, allowing visually impaired users to navigate efficiently and intuitively.

Website

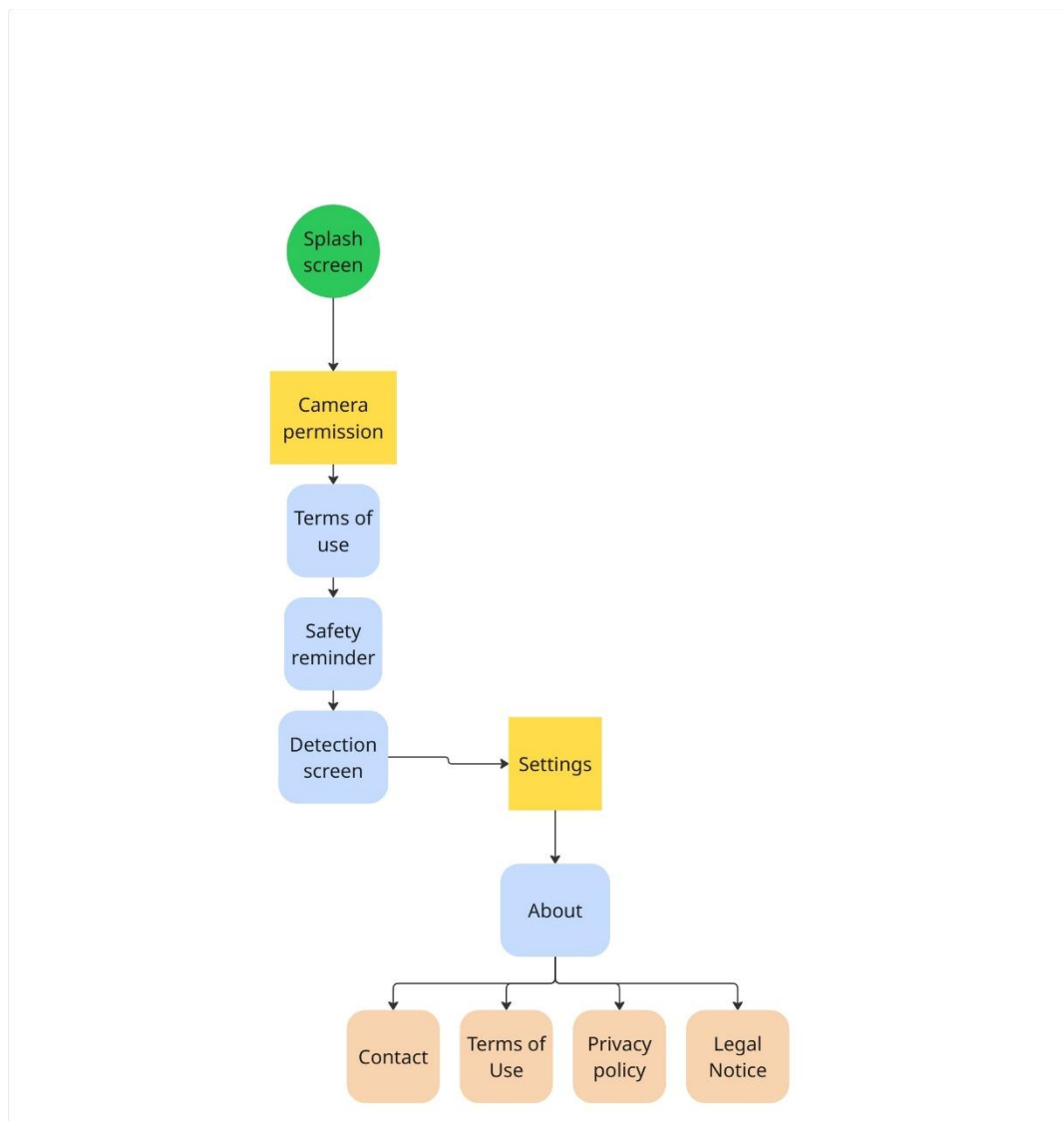


The Naviko website follows a single-page navigation approach to provide quick access to all essential information. Users can navigate between the different sections using the navigation menu or call-to-action buttons without leaving the page. Only the legal

documents (Terms of Use, Privacy Policy, and Legal Notice) are accessed separately.

Mobile Application

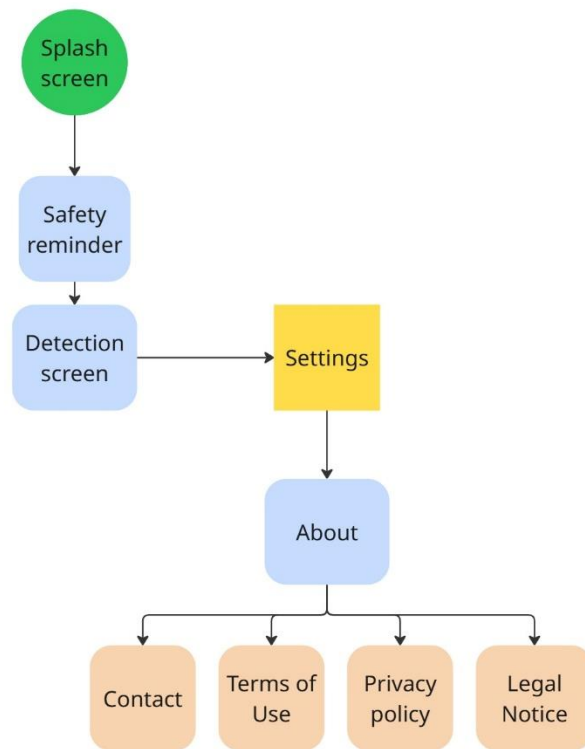
First launch



During the first launch, users must grant camera permission and accept the Terms of Use and Privacy Policy before accessing the application's detection features. Once completed, these steps are not required again unless the application is reinstalled or its data is reset.

Standard launch

Standard launch



On subsequent launches, the application displays a brief safety reminder before opening the real-time detection screen. Users can access personalization settings and legal information at any time through the Settings menu.

Budget and Schedule

Estimated Budget

Category	Estimated cost (€)
Domain name	15€/year
Hosting website	60€/year
Google play dev account	25€
Dev. software	0€
AI API	0~100€
Test device	Personal devices
Estimated total	100~200€

This budget is an estimate for a prototype version of Naviko. Development is carried out by a single developer using mainly free and open-source tools.

As this project is developed as part of a Bachelor's degree, development costs are not included in the estimated budget. Only external services and operational expenses are considered.

Project Timeline

Phase	Duration
Opportunity Study	Completed
Project Charter	Completed
Feasibility Study	Completed
Risk Analysis	Completed
Functional Analysis	Completed
Requirements Specification	In Progress
UI/UX Design	In Progress
Software Design	Planned
Development	Planned
First Prototype	Planned
Testing	Planned
Deployment	Planned

Gantt Summary

The project planning is summarized using a Gantt chart describing the main phases from project initiation to the development and testing stages. The schedule will be updated throughout the project as implementation progresses.

Functional Requirements

Application Launch

The application has been designed to provide quick and intuitive access to its main functionality while ensuring that users are informed about its limitations and safety recommendations.

- Display a splash screen when the application starts.
- Request camera permission during the first launch.
- Require users to accept the Terms of Use and Privacy Policy before using the application.
- Display a short safety reminder before opening the camera.
- Automatically launch the real-time detection screen.

Camera and Real-Time Detection

The primary function of Naviko is to analyze the user's surroundings in real time using the smartphone camera. The application highlights detected objects and provides immediate feedback.

- Display the live camera feed.
- Detect pedestrian crossings.
- Detect pedestrian traffic lights.
- Detect obstacles located in the user's path.
- Display detection boxes around recognized objects.
- Continuously update detections while the camera is active.

AI Features

Artificial Intelligence is responsible for recognizing objects captured by the smartphone camera. The system analyzes each frame and generates detection results in real time.

- Perform real-time object detection.
- Identify predefined objects relevant to pedestrian navigation.
- Display visual detection boxes.
- Provide voice and vibration feedback based on detected objects.

Accessibility Features

Accessibility is the core objective of Naviko. The interface has been designed to remain simple, readable, and adaptable to different levels of visual impairment.

- Allow users to adjust text size.
- Support high-contrast display.
- Provide voice alerts.
- Provide vibration feedback.
- Maintain a minimal and uncluttered interface.

Notifications and Alerts

Naviko informs users of detected objects through different feedback methods to improve awareness while minimizing distractions.

- Play voice alerts for important detections.
- Trigger vibration alerts when necessary.
- Prioritize critical detections.
- Avoid excessive or repetitive notifications.

Settings

The Settings menu allows users to personalize the application's accessibility options and access additional information.

- Modify text size.
- Enable or disable vibrations.
- Adjust contrast settings.
- Access the About section.
- Open the Terms of Use.
- Open the Privacy Policy.
- Open the Legal Notice.

Legal and Safety Requirements

As Naviko provides AI-assisted guidance, users must be informed about the application's limitations and responsibilities before use.

- Display safety information during the first launch.
- Inform users that AI detections may be inaccurate.
- Require acceptance of the Terms of Use before first use.
- Ensure legal documents remain accessible from the Settings menu.

User Stories

User stories describe the application's functional requirements from the perspective of the end user. They help define development priorities and guide the implementation process.

ID	User stories	Priority
US-00	As a visitor, I want to learn about Naviko's features	Medium

	and limitations before downloading the application.	
US-01	As a visually impaired user, I want Naviko to detect relevant elements in my environment so that I can move around more safely.	High
US-02	As a user, I want voice alerts to inform me about detected obstacles without looking at my phone.	High
US-03	As a user, I want to customize the application's visual settings so that I can adapt Naviko to my needs.	Medium
US-04	As a user, I want vibration alerts to receive feedback in noisy environments.	Medium
US-05	As a user, I want to access the Terms of Use and important safety information so that I understand the	Low

	conditions and limitations of using Naviko.	
--	---	--

Technical Requirements

Supported Platforms

Although React Native supports both Android and iOS, the first version of Naviko will be developed and tested exclusively on Android. This decision allows development efforts to focus on building a stable prototype before extending compatibility to additional platforms.

Hardware Requirements

Naviko relies on several smartphone hardware components to perform real-time environmental analysis.

- Camera (required)
- Speaker (required)
- Vibration motor (required)
- Internet connection (required for the first prototype)
- GPS (optional – reserved for a future version)

Software Requirements

The application is built using modern cross-platform and open-source technologies to ensure maintainability and scalability.

Component	Technology
Mobile Application	React Native
Programming Language	TypeScript
Backend API	Python
AI Processing	Python

Database	No need in V1
Version Control	Git/Github

React Native was selected because it allows the development of a cross-platform mobile application from a single codebase, reducing development time while maintaining the possibility of supporting both Android and iOS in future versions.

Python was chosen due to its extensive ecosystem for artificial intelligence, computer vision, and image processing, making it well suited for Naviko's detection features.

TypeScript improves code quality and maintainability through static typing, helping reduce errors during development.

Git and GitHub are used to manage source code, track changes, and maintain project documentation.

A database is not required for the first version since no user accounts or persistent data storage are implemented. PostgreSQL has been identified as a suitable solution for future versions.

Performance Requirements

The application should provide responsive interactions while maintaining acceptable performance on modern smartphones.

- Launch within a few seconds.
- Display camera preview smoothly.
- Perform real-time object detection.
- Minimize battery consumption.
- Minimize application crashes.

Security Requirements

User privacy and safety are essential aspects of Naviko.

- Request camera permission before use.
- Request user consent for Terms of Use.
- Do not collect personal information.
- Do not store camera images.
- Protect future communications using HTTPS.

Compatibility Requirements

Naviko should remain compatible with a wide range of smartphones and accessibility features.

- Support portrait orientation.
- Support Android accessibility features.
- Support different screen sizes.
- Maintain readability in dark mode.
- Adapt to different text sizes.

External Services and APIs

Naviko relies on external technologies to provide real-time AI-assisted object detection.

Service	Purpose
AI Detection Model	Object detection

Smartphone Camera API	Live camera access
Text-to-Speech API	Voice notification

Development Tools

Tool	Purpose
Visual Studio Code	Development
GitHub	Version Control
Figma	UI/UX Design
Trello	Project Management
Miro	Diagrams
Android Studio	Android Emulator & Testing

AI Requirements and Regulatory Constraints

AI Reliability

Naviko relies on artificial intelligence to detect objects in the user's environment. Although the system aims to provide accurate real-time assistance, AI predictions cannot be considered perfectly reliable.

- Perform real-time object detection.
- Display detections only when confidence is sufficient.
- Minimize false positives and false negatives.
- Clearly communicate that AI predictions may contain errors.
- Continuously improve detection accuracy during future development.

AI Limitations

The performance of the AI model depends on environmental conditions and cannot guarantee perfect object recognition in every situation.

- Low-light environments may reduce detection accuracy.
- Weather conditions (rain, fog, snow) may affect performance.
- Partial occlusions can prevent object recognition.
- Fast-moving objects may not always be detected.
- Camera quality influences detection performance.
- Unexpected objects may not be recognized.

User Safety

Naviko is designed as an assistive tool and must never replace the user's own judgment, mobility training, white cane, guide dog, or attention to the surrounding environment.

- Display a safety reminder before use.
- Inform users that AI detections are not guaranteed.
- Encourage users to remain attentive.
- Require acceptance of the Terms of Use.
- Never present the application as a substitute for mobility aids.

Data Protection (GDPR)

Naviko has been designed to minimize the collection of personal data and comply with the principles of the General Data Protection Regulation (GDPR).

- Do not require user registration.
- Do not store camera images.
- Do not collect unnecessary personal information.
- Inform users about camera usage.
- Provide an accessible Privacy Policy.

AI Act Considerations

Naviko has been designed with awareness of the European Union's AI Act. Although the project is currently developed as an academic prototype, its design takes into account the main principles promoted by the regulation, including transparency, user safety, and responsible use of artificial intelligence. Inform users that AI is used.

- Inform users whenever AI is used.
- Clearly communicate the limitations of AI-assisted detection.
- Keep the user in control of decision-making.
- Provide accessible safety information before use.
- Minimize the collection and processing of personal data.
- Consider future regulatory updates as the project evolves.

Ethical Considerations

Naviko has been developed to improve accessibility while respecting user autonomy, privacy, and safety.

- Accessibility is the primary objective.
- AI assists the user but does not replace decision-making.
- Respect user privacy.
- Avoid collecting unnecessary personal data.
- Clearly communicate AI limitations.
- Promote inclusive design.

Conclusion

Summary

This Software Requirements Specification defines the functional and technical requirements for Naviko, an AI-assisted mobile application designed to support visually impaired users during outdoor navigation.

The document presents the application's main features, technical architecture, accessibility objectives, AI-related considerations, and regulatory requirements. Particular attention has been given to user safety, transparency, and ease of use throughout the design process.

As an academic project, this specification serves as the foundation for the next stages of development and will guide future implementation and testing.

Next Project Phase

Following the completion of this Software Requirements Specification, the project will enter the software design phase.

The next stage includes:

- Defining the software architecture.
- Creating UML diagrams.
- Designing the database structure (if required in future versions).
- Preparing the project management workflow (Trello).
- Implementing the first application prototype.
- Performing functional and accessibility testing.

The objective of these next phases is to transform the documented requirements into a fully functional prototype while maintaining the project's accessibility and safety objectives.

Naviko aims to demonstrate how artificial intelligence can improve accessibility while remaining transparent, responsible, and centered on the needs of visually impaired users.